

Topic 8. ENERGY SOURCES.



DISCUSSION/SPEAKING PRACTICE

Renewable Energies and Energy Transition

Please study the following links.

TED Talks:

- http://www.ted.com/talks/donald_sadoway_the_missing_link_to_renewable_energy
- https://www.ted.com/talks/david_mackay_a_reality_check_on_renewables

Greta Thunberg on climate change

https://www.ted.com/talks/greta_thunberg_the_disarming_case_to_act_right_now_on_climate?language=de#t-19251

Articles:

- <https://www.energy.gov/eere/water/types-hydropower-plants>
- <https://www.cleanenergywire.org/factsheets/how-can-germany-keep-lights-renewable-energy-future>
- <http://www.dw.com/en/feldheim-germanys-renewable-village/a-18466800>
- <https://www.thesolutionsjournal.com/article/how-germany-became-europes-green-leader-a-look-at-four-decades-of-sustainable-policymaking/>
- <https://view.nl.npr.org/?qs=6efa7be2de6330734eddf55b075d198f5b7f7a63c5d9d5cf9be80dd2750c785022173160da1f2e8d9a6079ae5ef10ae4060112d3c3894431d736057005a274bedf8dc97b3f17afb8eff2b3ed839a0388a35ea95858c30c9f>

Statistics:

http://ec.europa.eu/eurostat/statistics-explained/index.php/Renewable_energy_statistics



In groups of 3 or 4 have a 20-minute discussion addressing all 4 agenda points mentioned in the setting. A representative of your group is going to summarize your arguments and conclusions and present them afterwards.

WATCH THE FOLLOWING VIDEO

https://www.youtube.com/watch?v=fyqDC_AKVgE (2019, 7 minutes)

Who is leading in renewable energy? | CNBC Explains

Answer the following questions:

1. In 2018 the world's energy generated from renewable sources was

- A. More than 20%
- B. Less than 25%
- C. More than 25%
- D. More than 50%

2. According to the speaker, what are the most common sources of renewable energy?

3. What are challenges of adopting renewables, e.g., solar?

4. Which two countries have the highest capacity to produce green energy and why?

5. What is Germany's situation as far as renewables?

Capacity including hydropower:

Capacity without hydropower:

Energy consumer:

Renewable power capacity per capita:

6. What is the most telling approach of looking at renewables in each particular country?

Which countries are at the forefront if this approach is applied?

7. Where in the world is the focus on renewables the least important?

8. What do you know about Kyoto Protocol and Paris climate accord?

9. How many countries have announced their unilateral plans of becoming climate-neutral by 2050?

10. Why was Elon Musk mentioned at the end of the video?

TOPIC RELATED VOCABULARY

<https://www.cleanenergywire.org/glossary>

This website offers a glossary on **energy transition**. Browse through it. Find 5-7 new terms connected with energy.

Be ready to present your selected terms. You start with the German equivalent/explanation, the class guesses.

READING I

KITE POWER

Read the text and complete the tasks. The words in *italics* are translated in German at the end of the text.

Part 1. POWER-GENERATING KITES IN THE CARIBBEAN

The island of Curacao bears all the hallmarks of a tropical paradise. A city-sized scrap of land in the Lesser Antilles, it boasts sandy beaches, coral reefs and an undersea cliff called "the blue edge" that attracts scuba divers from all over the world.

Look closer, though, and Curacao has its challenges. Curacao's 150,000 residents import a lot of essential goods, including the fossil fuels that generate much of their electricity. Dependence on imported fuel is ironic as well as expensive. Thanks to a huge oil refinery near its main harbour, Curacao had, in 2014, the second-highest carbon emissions per capita of any country in the world - not a great look for a low-lying island threatened by rising sea levels.

To Johannes Peschel, the answer to Curacao's carbon quandary lies in its greatest natural resource: the wind. Curacao was an early adopter of wind power, and by 2017 commercial wind farms were producing 30% of its energy. But Peschel, a German entrepreneur with a bushy beard and a surfer's laid-back enthusiasm, wants to push that figure higher, and his plans do not involve adding more turbines to the coastline. Instead, he and his local partners are pinning their hopes on a technology that is both centuries old and entirely new: giant, *tethered* kites that generate energy as they soar through the blue Caribbean sky.

Part 2. LET'S GO FLY A KITE

For Peschel and others in the small but growing airborne wind energy (AWE) community, the future of wind power looks nothing like the familiar three-bladed turbines scattered across the hills and coasts of Europe. Researchers in this nascent field are working on a dizzying array of devices, including kites, wings, drones and even a set of spinning sky-borne hoops that are being developed by a Scottish firm called Windswept and Interesting. In all cases, the goal is to harness high-altitude wind in a way that is cheaper and more flexible than erecting lofty columns of steel and concrete.

To residents of islands like Curacao, as well as communities in areas without conventional electrical grids, AWE offers substantial advantages. Peschel - whose company Kitepower spun out of a research group at TU Delft in the Netherlands - notes that its biggest kite, with an area of 100 m² and a generating capacity of 100 kW, fits into a large surf bag and can be launched by two people in 20 minutes. Fixed-wing AWE systems are bulkier, but not by much.

TwingTec, Kitepower and a clutch of similarly named companies like EnerKite, Kitemill, SkySails Power, Windlift and so on are pursuing the same path to market that conventional renewables followed a generation ago. By offering small, 10-100 kW systems to customers in remote locations - where costs per Kilowatt per hour are high and the main alternatives are the noisiest of diesel generators - they aim to refine their technology and prove its worth before scaling it up.

Other firms are targeting large-scale power generation from the start. A second Netherlands-based start-up, Ampyx Power, is developing a tethered aircraft that can take off and land on floating platforms far out to sea. Its newest model has a rated power of 300 kW, and chief executive Richard Ruiterkamp says the company will begin flying it at a purpose-built test site in County Mayo, Ireland, sometime next year. In August, the US-based firm Makani tested its 600 kW kite in a Norwegian fjord, collecting *reams* of data on how the kite-tether-buoy system behaves under real-world conditions. Both companies envision a future where large, multi-kite farms deliver hundreds of megawatts of low-carbon energy to the grid.

Part 3. UP TO THE HIGHEST HEIGHT

Kites have been used to pull loads since ancient times, but their power-generating potential was not fully appreciated until relatively recently. During the oil crisis of the late 1970s, Miles Loyd, an engineer at Lawrence Livermore National Laboratory in the US, began studying a phenomenon that novice kite-fliers learn the hard way: a kite moving perpendicular to the wind pulls much harder than a stationary kite with the wind behind it. This is because the kite's aerodynamic lift force scales with the square of its apparent airspeed. In cross-wind flight, this apparent airspeed can be easily an order of magnitude higher than the wind speed relative to the ground.

Loyd's contribution was to calculate that a kite flown in fast loops across the wind would produce enough lift not only to support itself, but also to generate a useful amount of power - hundreds of times more power, in fact, than a kite in static flight. Using realistic numbers for wing area, wind speed and *lift-to-drag ratio*, Loyd estimated that a single "crosswind" kite could produce as much as 45 MW. This was far higher than the turbines of Loyd's day, and it remains competitive even now. As of 2019, the world's most powerful commercial wind turbine, the Vestas V164, maxes out at 8.8 MW.

Falling oil prices took some of the shine off Loyd's ideas, and a few of his approximations would be right at home amid the frictionless surfaces and spherical cows of introductory physics courses. Even so, Loyd's work remains influential. Most AWE companies are building kites that fly across the wind and generate electricity using one of the two methods Loyd proposed.

If the power-generating potential of crosswind flying helped get AWE off the ground, other advantages may prove just as important in keeping it aloft. A key part of the technology's appeal is that kites can harvest wind energy that conventional turbines cannot. A typical 1.5 MW turbine stands 100 meters tall, and its net capacity factor- the actual energy produced, divided by the maximum possible - is usually less than 50% due to the intermittent nature of wind. Kites, in contrast, can fly at altitudes of up to 500 meters, where winds are stronger and more consistent.

The greatest attraction of AWE systems, though, is their low mass. An energy kite is a turbine stripped back to its power-generating essence, with no need for weighty components such as a base and a tower. Kitepower's co-founder Roland Schmehl, a researcher at TU Delft, compares it to a set of flying rotor tips. The drastic reduction in mass should, in principle, reduce costs, making wind energy more likely to be exploited.

AWE advocates say that their devices will make it possible to harness much more of the Earth's wind energy. A combined solar and AWE farm, where kites cast tiny, transient shadows as they circle high above the photovoltaic panels, is one possibility. Kitepower's

Curacao project, in which the company plans to operate three to five 100 kW kites on an existing conventional wind farm, is another. The most *tantalizing* prospect, though, is that AWE could unlock the “stranded resource” of wind in off-shore locations where the water is too deep to fix conventional turbine towers to the ocean floor. Compared to a floating turbine, a floating AWE platform would need a much less massive counterbalance to keep it upright in stormy seas.

Part 4. TURBULENCE AHEAD

All these advantages come at a price, and Ruiterkamp of Ampyx Power sums it up well. “The threshold for getting a first series to work is way higher than in other areas of wind energy,” he told his colleagues in Glasgow. “We are working on intrinsically unstable systems. If something goes wrong, we are immediately in a situation that needs to be handled.”

With a device as complex as an energy kite, the list of things that can go wrong is extensive. In another talk at the Glasgow meeting, Kitemill technical manager Lode Carnel rattled off a litany of problems that his Norway-based team had overcome on their current, *rigid-wing* prototype. Weak links breaking. Electrical connectors not up to industrial standards. Tether and wing materials that proved too flimsy or heavy.

It doesn't help that most AWE devices, unlike turbines, need to land when wind conditions are poor. Landings and launches are hard to automate, and each one raises the risk of catastrophic failure.

In off-shore operations, the bar for safe launch and landing is even higher. Each contributing system must work in a remote and harsh environment, with minimal maintenance and generally without supervision. In this respect, the industry's first offshore test was both an impressive milestone and a sign of how much work remains. Landing Makan's kite on a moving buoy was, McLeod observes, “like trying to parallel park a car while the kerb is moving up and down and back and forth, and also rotating for good measure”. Although that landing was successful, the test ended with the loss of the kite.

Part 5. LOOK OUT BELOW

In a young industry with a new technology, such mishaps are inevitable. They are also the reason why AWE is, at present, confined to lightly populated areas. At the Glasgow conference, there was broad agreement among delegates that an accident - such as a kite colliding with an aircraft, or with a person on the ground – would harm the entire sector, regardless of which company was responsible. But the question of how to regulate AWE provoked a rare public dispute, centred on a question that members of this friendly, freewheeling community sometimes struggled to answer: just what are these strange new objects in the sky?

For Neal Rickner, Makani's chief operations officer and a former F/A-18 pilot, the answer is simple. “We are talking about wind turbines, not aircraft,” he declared. “From a pilot's point of view, you would not expect an energy kite to be in one part of the sky at one moment, and another part at the next.” A failure of a kite's tether, Rickner argued, should be treated just like a failure in the blade of a conventional turbine.

Some of Rickner's fellow panellists were less convinced. Michiel Kruijff, head of technology at Ampyx Power, explained that its kite stores energy on board so it can fly back to its

platform if its tether breaks. That means it must follow aviation rules, although Kruijff said that Ampyx hopes to negotiate some adjustments. But in his view, even companies without these constraints need to face some facts. "We all acknowledge that we are primarily developing an energy-generation device," he said. "But we cannot deny that there is an element in it that is not like a wind turbine." Turning to Rickner, he asked rhetorically, "When you look at your beautiful system in the air, do you say, 'Look how beautifully it is rotating? Or do you say, 'Look how beautifully it is flying'?"

Part 6. FLYING HIGH AND GROWING BIG

Over the next few years, AWE developers hope to fly their devices for longer periods, with less human intervention. For companies at this stage of development, the *state-of-the-art* is days or weeks of automated flying. The long-term goal, especially for offshore applications, is to push that to months or even years of complete autonomy. It's an ambitious target, and it isn't clear when, if ever, it will be met.

Still, the physics behind AWE is sound, and it has a rare ability to capture the imagination. "People like kites, and they like green energy," says Joep Breuer, Kitepower's technical manager. "It's a very likeable technology." By 2027 the IEA predicts that wind will be the number-one source of the EU's energy, but even that will not be enough to meet *stringent* emissions targets. Alexander Bormann, the chief executive of Enerkite, *concur*s. "Some people think that airborne wind is crazy stuff," he says. "You know what I think is crazy? Installing less capacity in renewable energy. We need to save gigatonnes of carbon dioxide today, not tomorrow. We need to fly high and grow big."

Margaret Harris, INCH 24, #1 2020, pp. 6-11.

GLOSSARY

concur, to	zustimmen	reams	große Menge
freewheeling	hier: locker	rigid-wing	mit fester Tragfläche
inevitable	unvermeidbar	state-of-the-art	Stand der Technik
intrinsically	in sich, eigentlich, wesenhaft	stringent	streng, strikt
kerb BrE/curb AmE	Bordstein	tantalizing	verlockend, spannend
lift-to-drag ratio	Gleitzahl	tether	Halteseil, Leine

Task 1.

1. Why does the author state that dependence on imported fuel in Curacao is "ironic"?

2. What happened after the price of oil dropped?
 - a. Loyd's ideas were still considered relevant
 - b. His work was largely forgotten
 - c. His methods were never used again

3. What is the strongest argument for wind kites?

- a. low weight
- b. easy set-up
- c. safety issues

4. Two of the main problems of wind kites are:

- a. the risk involved if technical problems occur
- b. difficulty in attaching the towers to the ocean bed
- c. the necessity to land when wind conditions are poor
- d. the cost
- e. acceptance by the general public

5. Why is it important to define correctly what a power-generating kite is?

Task 2. Match the words from the text with words with similar meanings:

1. hallmark (part 1)	a. the most typical quality of feature
2. quandary (part 1)	b. strict, severe, harsh
3. early adopter (part 1)	c. arrangement, assemblage
4. to tether (part 1)	d. unwieldy, cumbersome
5. nascent (part 2)	e. beginner, neophyte, newcomer
6. array (part 2)	f. agree, be in accordance
7. bulky (part 2)	g. dilemma, plight, predicament
8. novice (part 3)	h. attach with a rope or chain so that it can't move
9. stringent (part 6)	i. a person who starts using a product/technology as soon as possible
10. concur (part 6)	j. developing, emergent, growing

READING II

MOON POWER

Read the text and complete the tasks. There is a glossary at the end of each part of the text.

Not only the sun provides us with free power but the moon does as well. Tidal power, usually dismissed as somewhat exotic, is finally coming of age: Orbital Marine Power has just launched the world's most powerful tidal turbine, the two-megawatt, floating O2.

When we talk about renewable energies, we usually refer to harnessing the sun's inexhaustible energy via photovoltaics, solar thermal systems or, somewhat indirectly, wind power. But there's another celestial body providing us with free energy. One capable of lifting 1.4 billion cubic kilometres of water twice a day: the Moon. As powerful and reliable as the tides are, they strangely play only a minor role in today's renewable energy mix. Something Orbital Marine Power wants to change.

The Scottish-based company has just anchored the world's most powerful tidal turbine at a test site near the Orkney Islands. With its slender body and two stubby 'wings' for the generator nacelles and rotors, the O2 looks out of this world - almost like a space ship, which is actually quite fitting for something that is capable of drawing two megawatts of power from the Moon. With such a capacity the O2 might finally "turn the tide" and convert ocean currents into a viable energy source.

Tapping the tides for some clean energy may seem obvious but so far the development of the necessary technology has been little more than a neap tide. One of the earliest and probably best-known projects is the Rance tidal power station in France, opened in 1966. It cuts off the estuary of the Rance River with a barrage to form an artificial tidal pool. The water flowing in and out drives 24 turbines which produce a peak power of 240 megawatts. While the construction of tidal barrages is relatively simple, suitable natural estuaries with a large tidal range are rare and damming them off can cause environmental issues. Therefore, tidal power was always considered an oddity - until someone had the idea to sink a wind power plant.

GLOSSARY

barrage	Wehr, Staudamm	harness, to	nutzbar machen, erschließen
celestial body	Himmelskörper	inexhaustible	unerschöpflich
come of age, to	erwachsen werden	nacelle	Gondel, Maschinenhaus
current	Strömung, Strom	neap tide	Nippflut
dam, to	eindämmen, aufstauen	oddity	Kuriosität, seltene Erscheinung
estuary	Mündungsgebiet	stubby	gedrungen, kurz und dick

UNDERWATER WINDMILLS

Dipping a wind mill into water actually makes a lot of sense. Seawater is almost 800 times as dense as air and also incompressible. As a result, the kinetic energy from a 5-knot (2.6 m/s) ocean current is equivalent to that of a major hurricane with a wind speed of 270 kilometres per hour. In other words, to produce one megawatt, a wind power plant needs a 55-meter rotor while a tidal turbine can do with a 20-meter rotor. Although intermittent, the tides are also highly predictable unlike the wind. Admittedly, there are fewer suitable locations for tidal turbines than for offshore wind turbines but tidal currents are nonetheless a force to be reckoned with.

The British government was one of the first to take this force seriously. At the beginning of the millennium, it recommended setting up a wave and tidal energy test facility to kick start a marine energy industry in the UK. A reasonable proposition for a nation that is basically an island surrounded by islands with many suitable sites. In 2003, the EMEC, or European Marine Energy Centre Ltd, was established. Today, it operates several test sites on the Orkney Islands and supports tidal energy companies in testing and commercialising their turbines. As a result, the UK has become the "Silicon Valley of tidal turbines". Its tidal power resource is estimated to be more than 10 gigawatts, representing about 50 percent of Europe's total tidal energy capacity. Moon power has become so ubiquitous in the British Isles that in the Shetlands you can turn your electric vehicle into a *moon buggy* by charging it at a tidal-powered charge point. And the distilleries in Islay and Jura will soon be powered by the tides as well giving a whole new meaning to the word '*moonshine*'.

One of the pioneers of tidal turbines was Marine Current Turbines. Founded in 2000 by marine engineer Peter Fraenkel, the company developed several prototypes, which, in 2008, culminated in the Sea Gen, the world's first large-scale commercial tidal stream generator. Its design was somewhat similar to the O2 as it also had a "wing" with two nacelles and two-bladed rotors which generated 1.2 MW for between 18 and 20 hours a day. The Sea Gen, however, was *mounted* on a pillar *secured* to the seabed on which it could travel up and down for maintenance. The project was highly successful and fed power into the grid of Northern Ireland until 2016 when it was *decommissioned*.

It's early days for the tidal power industry, however, and not unlike at the beginning of the wind power industry, the business is highly *volatile*. Peter Fraenkel turned his attention to gravity storage and founded Gravitricity while Marine Current Turbines intellectual property was, after a few mergers, dissolved into Simec Atlantis Energy, one of the big players in tidal energy. Atlantis is currently working on sinking dozens of their AR 1500 turbines between the Orkneys and the Scottish mainland to form the world's largest tidal energy plant.

The AR 1500 is a tidal turbine with a single nacelle and a three-bladed, 10-meter rotor that sits on a stand on the seafloor. The 1.5-megawatt turbine is permanently submerged, held in place by a heavy foundation.

GLOSSARY

a force to be reckoned with	Ein ernstzunehmender Faktor	grid	(Strom-) Netz
admittedly	zugegebenermaßen	moon buggy	Mondauto
charge point	Ladestation	moonshine	hier: schwarzgebrannter Schnaps
currently	aktuell	ubiquitous	allgegenwärtig, flächendeckend
decommission, to	außer Betrieb nehmen, stilllegen	volatile	unbeständig, schnelllebig
gravity storage	Schwerkraftspeicher		

SINK OR SWIM

While most competitors follow a similar approach - a single rotor turbine anchored to the seafloor - the jury is still out whether this "drop and forget" strategy will work. For any maintenance the whole structure has to be recovered, which requires specialised and expensive heavy lift vessels. This drives up maintenance costs, which play a critical role in making tidal power a viable source for renewable energy due to the turbines' long life cycle. Designing the system as maintenance free as possible and opting for expensive but

infrequent repairs is one way of tackling the problem. The other is to build a space ship - or a moon ship to be precise.

Orbital Marine Power's O2 is a floating tidal turbine. Their "moon ship" has a 72-meter welded steel *tubular hull*, which acts as a floating platform and houses the hydraulics and other subsystems. Attached to the hull are two 18-meter-long articulating arms carrying the two nacelles. Each nacelle consists of a one-Megawatt generator and a 20-meter rotor with two blades which can be *pitch* controlled around 360 degrees to catch incoming as well as outgoing tides. During normal operation the arms are angled about 50 degrees downwards so that the rotors are submerged. For maintenance they can be lifted into a horizontal position allowing access to the arms, the nacelles and the rotor's hub via *hatches*.

It's this easy access which Orbital's CEO, Andrew Scott, thinks will give the O2 the commercial edge: "Alternative technologies with similar installed capacity fixed to the seabed are locked into using expensive vessels and equipment to recover them as they can't be accessed while they're fixed to the seabed. Our costs are a *fraction* of our competitors', and ultimately a totally different risk profile for investors and the supply chain." After all, the O2 can be *towed* like a ship to its operation site where it's *moored* to the ocean floor at four points with minimal undersea work. Regular maintenance can be carried out with small, low-cost workboats.

Critics may argue that the floating tubes will *obstruct* the shipping traffic but so do offshore wind parks. Harnessing tidal currents close to the surface has the additional advantage that the flow is generally more energetic and not as turbulent as close to the seabed.

Orbital is confident that the O2 will 'turn the tide' on commercialisation and they have every reason to be. The O2's experimental predecessor, the similarly shaped 2MW SR1-2000, fed into the grid of the Orkneys from the start of 2017 until August 2018, meeting about 25 percent of the electricity needs of the islands during that period. With the tail wind or, in this case, "incoming tide" of a fundraising and crowdfunding campaign, Orbital was able to build the O2, basically an *overhauled* SR1-2000 to drive down costs.

GLOSSARY

edge	hier: Vorteil	heavy lift vessel	Schwergutschiff, Kranschiff
estimate, to	schätzen, veranschlagen	hub	Nabe
feed, to (fed, fed)	hier: einspeisen	pitch	Neigung, Blattverstellung
found, to	gründen, etablieren	the jury is still out	es ist noch nicht entschieden
foundation	Fundament, Unterbau	tubular	röhrenförmig
hatch	Luke, Klappe	two-bladed	zweiblättrig

LEARNING FROM THE WIND

A key factor in that process was taking advantage of the knowledge and components of the already matured wind energy industry. After all, a tidal turbine is essentially an underwater windmill.

Consequently, Orbital contracted SKF, a long-standing supplier to the wind power industry, to deliver the complete, fully integrated *drive train* as a 'plug & play module'. SKF's contribution, however, was more than just *bearings* and *seals*. Their pitch-control modules,

which rotate the blades around their longitudinal axis, constantly regulate the *angle of attack* to optimise and control the power output. The enhanced blade control also smooths out power fluctuations, which reduces extreme loading and may even allow the operators to increase the rotor diameter in the future - just as it did in wind power plants.

But the analogy only carries so far. "The real difference is around what you mount your nacelle to, and in that context our floating platform serves the same function as a wind turbine tower," says Scott. It was probably a matter of national pride that this "tower" was built and the whole structure assembled in Scotland. On the 22 April, the O2 left the Port of Dundee on a *submersible barge*, making it the first vessel to launch from the town since ship building ended over forty years ago. After arrival in Orkney a couple of days later, the final commissioning at EMEC's Fall of Warress tidal test site is currently under way.

The future will tell whether the O2 will be even more successful than its predecessor, whether floating or fixed turbines are more economical or, for that matter, whether tidal power will finally see its spring tide.

GLOSSARY

angle of attack	Anströmwinkel, Anstellwinkel	obstruct, to	blockieren
bearing	Lager	overhaul, to	überarbeiten, überholen
drive train	Antriebsstrang	predecessor	Vorgänger-/in
fraction	Bruchteil	seal	Dichtung
fundraising	Finanzmittelbeschaffung	submersible barge	Halbtaucher-Barge
moor, to	Festmachen, vertäuen	tail wind	Rückenwind

Matthias Meier, INCH 29, #2 2021, pp. 7-11.

1. Why does the author feel that it's "strange" that tides play such a small role in the energy sector?

2. What is one main benefit of using the tides for power generation?

- a. Little negative impact on estuary ecosystem
- b. There are many appropriate locations
- c. The predictability of tides

3. Give 2 reasons why the UK is considered a "natural setting" for tidal power.

4. Why was Sea Gen decommissioned?

- a. It was too expensive
- b. There were many maintenance problems

c. Not given

5. What is the “drop and forget” strategy?

- a. The turbines are abandoned at sea when they no longer function
- b. The turbines require no maintenance
- c. The turbines are attached to the seafloor

READING III

BATTERIES OF THE FUTURE, SODIUM OR LITHIUM?

Read the text and complete the tasks.

Introduction

Sodium-ion rechargeable batteries could soon be a cheaper and resource-saving alternative to current lithium-ion cells. Powerful prototypes and groundbreaking findings in basic research make a breakthrough seem imminent. Electro mobility, especially in combination with renewable electricity production, is regarded worldwide as a climate-friendly model for the future. However, a decade-long focus on the use of the rare and expensive metal lithium in rechargeable batteries has slowed down the large-scale electrification of vehicles across the world.

Although electric cars, commercial vehicles and e-bikes are gaining ground, their limited ranges, high prices, a cumbersome charging infrastructure and resource-intensive battery production are delaying a rapid turnaround. For electric vehicles to come into more widespread use, batteries must become even more powerful, durable, sustainable and cost-effective. The solution could be sodium-ion batteries, whose development has recently made astonishing progress. In the foreseeable future, they could replace the lithium-ion batteries currently used not only in electric vehicles, but also in smartphones and laptops.

The two alkali metals lithium and sodium are chemically very similar. Although sodium does not have the energy density of the comparatively rare lithium, it is widely and cheaply available.

Part 1. BREAKTHROUGH IN SIGHT

Currently, however, the performance of sodium-ion batteries lags behind that of lithium-ion batteries by about 20 years. For decades, research has been concentrated solely on the more powerful lithium. Now however, there are not only ground-breaking scientific publications, but also very promising prototypes.

A South Korean sodium-ion battery managed to handle about 500 complete charging cycles before its capacity dropped to 80 %, according to a May 2020 publication.

A battery with a slightly different chemical structure devised by a US-Chinese research group achieved 450 charge cycles with a similar charging capacity. And a Chinese sodium-ion battery had a slightly lower capacity, but still retained 70 % of its capacity after 1,200 cycles of quick 12-minute charging.

All this may not sound like much, but in practice these batteries would probably survive many more charge cycles, because in everyday life batteries tend to be only partially charged and discharged. The complete loading and unloading of a battery in an experiment puts much more strain on the cell. In addition, sodium-ion technology does not consume any scarce resources. The production of the cathodes does not require rare lithium salts; simple table salt is sufficient. Powerful anodes can be produced from lignite, wood and other biomass. Cobalt or similar rare substances are also not required.

Part 2. MULTILAYER SOLUTION

However, sodium has two disadvantages. For one, it is three times heavier than lithium, so sodium-ion batteries are also heavier, even though lithium accounts for less than 5% of the total weight of a battery. In addition, sodium batteries are less powerful because they inevitably lose around 10% of their energy density due to a 0.3-volt lower cell voltage. This is largely owing to the fact that the graphite anodes that have been used up to now in batteries absorb too little sodium.

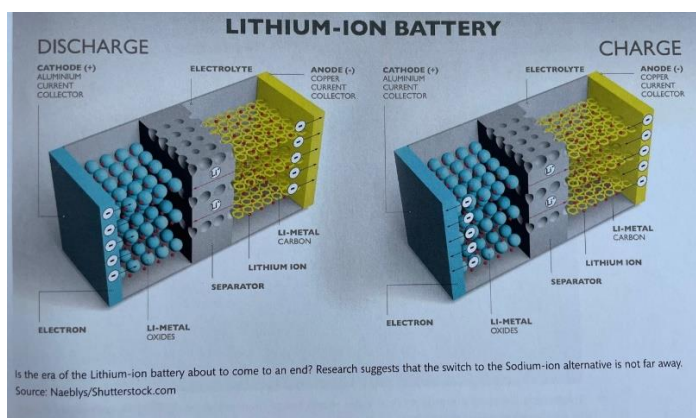
But nanoscale carbon could provide a remedy. This is shown in a study by a German-Russian working group led by the Helmholtz Centre Dresden-Rossendorf (HDR). The study suggests that double layers of graphene, i.e., wafer-thin carbon, could store significantly more sodium atoms in the anode than the graphite used so far.

Part 3. POSSIBLE NEW MOMENTUM

If multilayer storage works for lithium, could it also work for other alkali metals such as sodium? The research group from Dresden, Stuttgart and Moscow looked into this question using complex supercomputer simulations. And their calculations have shown that sodium, like lithium, can be deposited between the graphene sheets not only as a single layer, but in several layers one on top of the other.

If graphene electrodes were to be installed in sodium-ion batteries in the future instead of the graphite anodes commonly used today, it might also be possible to achieve a significantly higher storage capacity.

These results could be a breakthrough for the development of future anodes for low-cost sodium batteries. "Our work is of a purely theoretical nature, and we do not claim that a new generation of batteries will be developed in the foreseeable future on the basis of our results," Krashenninnikov emphasizes. "But perhaps our findings will give engineers some new and interesting ideas."



Part 4. THE WHITE GOLD OF E-MOBILITY

A small community in the German state of Saxony may soon see the revival of its centuries-old mining tradition. But this time around, the focus will no longer be on tin or tungsten but on lithium. The Saxon community of Zinnwald-Georgenfeld near the German-Czech border, some 50 kilometers (31 miles) south of Dresden happens to be in an area that has attracted the interest of geologists and mining companies. Lithium, the new white gold of the automotive and other industries abounds in Zinnwald-Georgenfeld.

As mentioned above, lithium is the chemical element needed for batteries in electric cars, but is also required in steadily rising quantities for glass ceramics, ceramic glass cooktops and a number of lubricants, to name but a few fields of application. Armin Müller, CEO of German mining company Deutsche Lithium Ltd. is confident that now that a feasibility study has been presented, financing details for the lithium project can follow in due course. If everything goes according to plan, Zinnwald could see the revival of its roughly 750-year-old mining tradition. "At the moment, we plan to mine 522,000 tons of lithium-containing ore per year," said Müller. "We'll use the material to get 125,000 tons of zinnwaldite concentrate per annum. This will then be processed to extract 5,100 tons of lithium fluoride annually. There is enough in the ground to allow us to do this for 30 years which is also the period for which we have a mining license."

Even though Zinnwald does not boast the amount of lithium in the ground that the world's largest exporters have at their disposal, growing demand worldwide speaks in favor of a successful domestic initiative, says Michael Schmidt, a geologist and senior analyst from the Berlin-based German Mineral Resources Agency (DERA) at the Federal Institute for Geosciences and Natural Resources. "Demand for lithium has been rising exponentially lately, particularly in the past 3 to 4 years because of the global e-mobility boom and increasing interest in renewable energy, like solar and wind," says Schmidt. "Back in 2015, global demand for lithium metal content exceeded 30,000 tons, or 150,000 tons of lithium carbonate, to use the more common compound. If you look at the current projections, the world will need between 600,000 tons to 1 million tons of lithium carbonate annually by 2025, depending on the development of e-mobility technology."

So far, Germany has been importing the raw material from remote regions of Chile, Argentina and Australia. While imports will still be needed in the future, and are also expected to come from Bolivia soon, the Zinnwald project, as well as other areas across Europe, could help ease the pressure, Schmidt argues. "If a company here runs the mining business and controls production costs in a way that enables it to stand its ground on the global market, it has every right to exist," he said. "It would not be a bad thing, given the oligopolistic lithium market right now - three nations control some 90% of lithium mining. This tells us that diversification would make perfect sense for industry and in terms of pricing." Schmidt's view is echoed by Doris Schüler from the Freiburg-based non-profit Institute for Applied Ecology. "The exploitation of deposits within the European Union, particularly lithium, nickel and cobalt, should be supported," she said. "Considerable imports would nevertheless be required to meet primary demand in the bloc. But those imports could be reduced, and mining in the EU would go hand in hand with regulations to protect the environment."

The zinnwaldite mineral only contains about 1.6% lithium. That compares poorly to spodumene in Australia, which contains about twice as much, meaning that twice as much ore will have to be mined in Zinnwald to get the same amount of lithium in the end. This

makes operations here more expensive, but Deutsche Lithium CEO Armin Müller points to a decisive cost-saving factor that most others do not have. "More often than not, lithium deposits are located in the middle of nowhere, somewhere in the middle of a desert or high up in remote mountainous regions," Müller said. "This means you have to spend a lot of resources on getting long transport routes to faraway chemical processing plants up and working. Such plants can mostly be found in Asia where lithium carbonate is processed," he explained.

"Now, we don't have to produce lithium carbonate at all, we don't need this interim procedure as we can directly process our mineral concentrate and make products such as lithium fluoride or other industry-ready compounds. This is because we can use a chemical plant that will be just around the corner in a place called Freiberg, which is only 49 kilometers (30 miles) away from the deposit."

Geologist Michael Schmidt says that analyzing the impact of e-mobility on the environment is important. "Short transport routes are essential," he said while talking about lithium mining in general. "After all, the initial idea behind e-mobility was to reduce the CO₂ footprint, meaning it would of course make sense to decrease that footprint also during the production of electric cars. If battery cells are to be produced in Europe, it makes sense to get the required resources from places nearby."

"After all, we're not talking about open-cast mining here. Yes, there will be a new building nearby in Altenberg where the stones from the mine will be pulverized. Trucks will then carry the material to places further away, but that's the only way tourists would know that mining is going on here at all." Armin Müller sees it exactly the same way. He is keeping his fingers crossed that outstanding financing and planning issues can be resolved in due course.

Part 5. SODIUM VS. LITHIUM PERFORMANCE

An important question to answer for this discussion is "how do Na-ion batteries compare to their Li-ion counterparts" in terms of energy densities, cost, applications, and overall consumer acceptance.

The foremost advantage of Na-ion batteries comes from the natural abundance and lower cost of sodium compared with lithium. The abundance of Na to Li in the earth's crust is 23600 ppm to 20 ppm (ppm = parts per million or mg/L), and the overall cost of extraction and purification of Na is less than that of Li. Moreover, Na-containing metal oxide and polyanion cathode materials can be fabricated from naturally abundant transition metals such as iron, manganese, vanadium, and titanium, without using cobalt, making Na-ion batteries sustainable and affordable in rich and poor countries alike.

Avicenne, a French battery market forecast company, has reported that the worldwide Li-ion battery market will grow to more than \$150 billion by 2025. The ESS market alone is expected to be >\$50 billion. The higher volume of battery materials production in response to this rising market is expected to reduce the cost of Li-ion batteries at the pack level to about \$100/kWh from the present \$150/kWh.

According to Avicenne, the cathode is the costliest component of a Li-ion battery at about 25% of the total cost. An examination of Li-ion and Na-ion battery components reveals that the nature of the cathode material is the main difference between the two batteries. Because the preparation cost of the cathode from raw materials is more or less the same for both Li-ion and Na-ion technologies, the main cost reduction for Na-ion batteries comes

from raw materials. On the basis of the information currently available we can project the cost of Na-ion batteries to be about 10-20% less than that of their Li-ion counterparts.

The major advantage of Na-ion batteries is sustainability, which is important for a world striving to be free of carbon-based energy sources. We can foresee Na-ion batteries with hard-carbon anodes and cobalt-free cathodes as sustainable lower-cost alternatives to Li-ion batteries for applications such as short-range electric vehicles and large-scale energy storage (ESS) in a world that is increasingly being transformed to wind, solar, and hydroelectric power, which depend on battery energy storage for uninterrupted, around-the-clock, performance.

Future research should focus on discovering advanced anode and cathode materials for Na-ion batteries with higher specific capacities and voltages so as to produce practical Na-ion batteries with specific energies approaching 200 Wh/kg. Efforts should also be made to develop advanced electrolytes that enable Na-ion battery performance at high charge-discharge rates over a wide temperature range while exhibiting the long cycle-life and shelf-life required for large-scale energy storage applications. Research and development efforts should also continue on Na-ion battery prototypes with particular emphasis on evaluating their temperature and rate-dependent performance and safety hazards.

Engine, #1 2021, pp. 28-33.

1. What issues hinder the overall reliance on e-mobility? Name 4.

2. Why has research always concentrated on the lithium battery?

- a. Lithium was widely available
- b. It was heavily promoted by the Chinese government
- c. Not given

3. Sodium batteries are likely to become more prevalent within..

- a. 5 years
- b. 10-15 years
- c. Not given

4. What three positive benefits are there for mining lithium in the Zinnwald?

5. How do Na-ion batteries compare to their Li-ion counterparts?

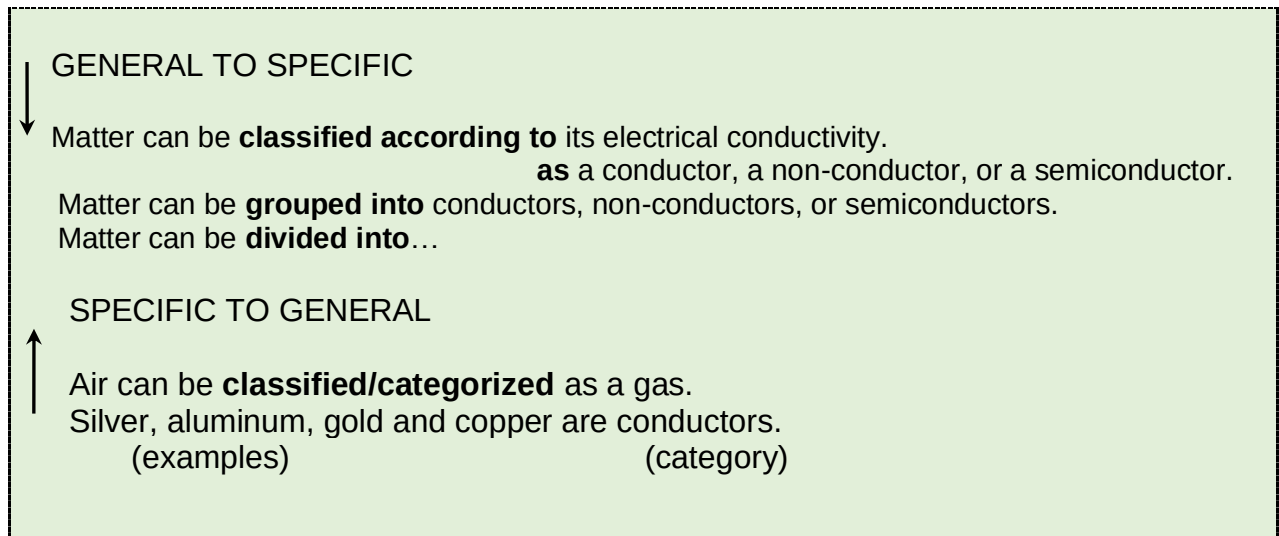
Na-ion batteries vs Li-ion batteries

Task 2. Match the words from the text with words with similar meanings:

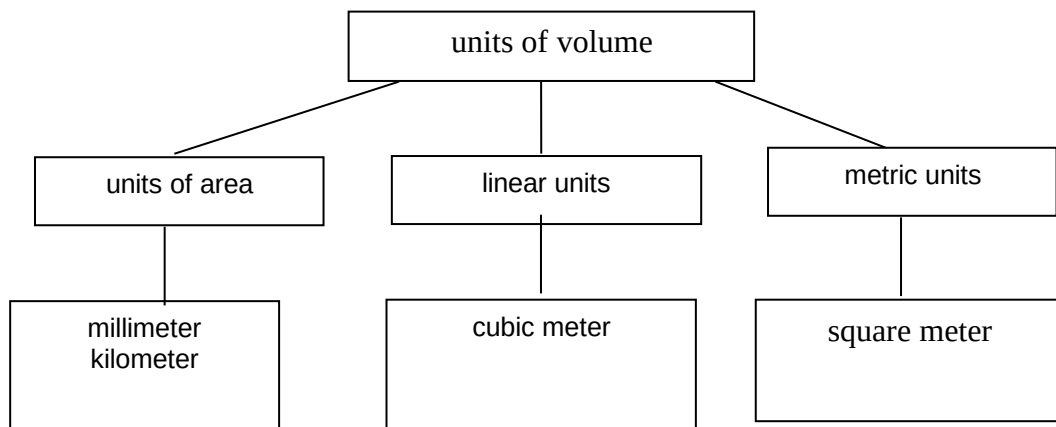
1. imminent (Introduction)	A intermediate
2. cumbersome (Introduction)	B close, near, approaching
3. alkali (Introduction)	C crushed, ground
4. lignite (Part 1)	D graphene
5. nanoscale carbon (Parts 2, 3)	E a substance with a pH value of more than 7
6. revival (Part 4)	F a soft brownish coal showing traces of plant structure, brown coal
7. to boast (Part 4)	G improvement, rally, upswing
8. remote (Part 4)	H unmanageable, awkward
9. interim (Part 4)	I possess, have, own
10. pulverized (Part 4)	G faraway, distant, secluded

Classification/Taxonomy

The word 'classification' comes from the word 'class', meaning a group of things that all have one important element in common. When you classify substances, objects, etc., you arrange or group them according to some system or principle.



Exercise 1. Correct the classification diagram where necessary.



Exercise 2. Create a classification diagram using the words in the box, add any other elements. Describe your diagram using the energy sources vocabulary.

solar...non-renewables...energy sources...renewables...
wind...geothermal...gas...hard coal
